



Sri Chaitanya IIT Academy.,India.

A.P. T.S. KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT

JEE-MAIN

Date: 31-12-2025

Time: 09.00Am to 12.00Pm

GTM-07

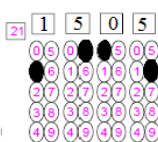
Max. Marks: 300

IMPORTANT INSTRUCTION:

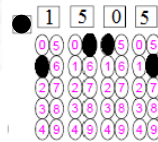
1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
 2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
 3. The test is of **3 hours** duration.
 4. The Test Booklet consists of **75 Questions**. The maximum marks are **300**.
 5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics and Chemistry** having **25 Questions** in each subject and subject having **two sections**.
(I) Section-I contains **20 Multiple Choice Questions** with only one correct option.
Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
(II) Section-II contains **05 Numerical Value Type Questions**.
- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking



Question Cancelled for Marking

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

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**MATHEMATICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

1. Let ABC be a triangle. Let A be the point (1,2), $y = x$ is the perpendicular bisector of AB and $x - 2y + 1 = 0$ is the angle bisector of $\angle C$. If equation of BC is given by $ax - by - 5 = 0$, then the value of $|a - b|$ is _____ where a and b are coprime, $a, b \in \mathbb{N}$

1) 1 2) 2 3) 3 4) 4

2. The point $(-2m, m+1)$ is an interior point of the smaller region bounded by the circle $x^2 + y^2 = 4$ and the parabola $y^2 = 4x$. Then m belongs to the interval

1) $-5 - 2\sqrt{6} < m < -1$ 2) $0 < m < 4$ 3) $-5 + 2\sqrt{6} < m < \frac{3}{5}$ 4) $-1 < m < -5 + 2\sqrt{6}$

3. Choose the correct matching from list I to II from the codes given below.

List-I		List-II	
(p)	$f(x) = x^3 $	(1)	continuous in $(-1,1)$
(q)	$f(x) = \sqrt{ x }$ is	(2)	discontinuous in $(-1,1)$
(r)	$f(x) = \sin^{-1} x $ is	(3)	differentiable in $(0,1)$
(s)	$f(x) = \cos^{-1} x $ is	(4)	not differentiable in atleast at one point in $(-1,1)$
		(5)	differentiable in $(-1,1)$

Codes:

1) p - 1, 3, 5; q - 2, 4; r - 1, 3, 4; s - 1, 3, 5 2) p - 2, 4; q - 1, 3, 5; r - 1, 3, 5; s - 1, 3, 4
3) p - 1, 3, 4; q - 1, 3, 4; r - 2, 4; s - 2, 4 4) p - 1, 3, 5; q - 1, 3, 4; r - 1, 3, 4; s - 1, 3, 4

4. If A and B are square matrices of order $n \times n$ such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will always be true?

1) $A = B$ 2) $AB = BA$
3) either of A or B is a zero matrix 4) either of A or B is an identity matrix

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5. The greatest value of the term independent of x in the expansion of $(x \sin \alpha + x^{-1} \cos \alpha)^{10}, \alpha \in \mathbb{R}$, is
- 1) $\frac{10!}{2^5}$ 2) $\frac{10!}{(5!)^2}$ 3) $\frac{1}{2^5} \frac{10!}{(5!)^2}$ 4) 1
6. For $x \in \mathbb{R}, f(x) = |\log_{10} 2 - \sin x|$ and $g(x) = f \circ f(x)$, then:
- 1) g is differentiable at $x = 0$ and $g'(0) = -\sin(\log_{10} 2)$
 2) g is not differentiable at $x = 0$
 3) $g'(0) = \cos(\log_{10} 2)$
 4) $g'(0) = -\cos(\log_{10} 2)$
7. Let $f(x)$ and $g(x)$ are two differentiable functions satisfying $f(x) + 3g(x) = x^2 + x + 6$,
 $f(x) + 2g(x) = x^2 + 2$ If $J = \int_0^{\pi/4} \ln(g(\tan^2 x) - f(\tan x) - 8) dx$, then find the value of $\frac{3\pi \ln 2}{J}$
- 1) 6 2) 7 3) 8 4) 9
8. Consider two curves $C_1 : y^2 = 4[\sqrt{y}]x$ and $C_2 : x^2 = 4[\sqrt{x}]y$, where $[\cdot]$ denotes the greatest integer function. The area of region enclosed by these two curves within the square formed by the lines $x = 1, y = 1, x = 4, y = 4$ is
- 1) $\frac{8}{3}$ sq.units 2) $\frac{10}{3}$ sq.units 3) $\frac{11}{3}$ sq.units 4) $\frac{11}{4}$ sq.units
9. The orthogonal trajectory of system of curve $y = ax^2$ ($a \neq 0$) which does not pass through origin, is
- 1) Ellipse 2) Parabola 3) Circle 4) Hyperbola
10. If α and β are the roots of the equation $x^2 - p(x+1) - q = 0$, then the value of $\frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + q} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + q}$ is equal to:
- 1) 1 2) 2 3) 3 4) 4





11. Let S be the set of all triangles in the xy-plane, each having one vertex at the origin and the other two vertices lie on coordinate axes with integral coordinates. If each triangle in S has area 50sq.units, then the number of elements in the set S is:

1)32 2)36 3)9 4)18

12. If S is the set of distinct values of 'b' for which the system of linear equations $x + y + z = 1$, $x + ay + z = 1$, $ax + by + z = 0$ has no solution, then S is

1) an infinite set 2) a finite set containing two or more elements
3) a singleton set 4) an empty set

13. Let $a_1, a_2, a_3, \dots, a_{10}$ be in G.P. with $a_i > 0$ for $i = 1, 2, 3, \dots, 10$ and S be the set of pairs (r, k) , $r, k \in \mathbb{N}$ (the set of natural numbers) for which

$$\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0.$$
 Then the number of elements in S, is

1)10 2)4 3)2 4) infinitely many

14. **Statement-1:** The system of linear equations $x + (\sin \alpha)y + (\cos \alpha)z = 0$;

$x + (\cos \alpha)y + (\sin \alpha)z = 0$ and $x - (\sin \alpha)y + (\cos \alpha)z = 0$ has a non-trivial solution for only one value of α belonging to the interval $\left(0, \frac{\pi}{2}\right)$

Statement-2: The equation in α given by $\begin{vmatrix} \cos \alpha & \sin \alpha & \cos \alpha \\ \sin \alpha & \cos \alpha & \sin \alpha \\ \cos \alpha & -\sin \alpha & -\cos \alpha \end{vmatrix} = 0$ has two solutions lying

in the interval $\left(0, \frac{\pi}{2}\right)$

- 1) Statement-1 is true, Statement-2 is true.
2) Statement-1 is false, Statement-2 is true
3) Statement-1 is true, Statement-2 is false.
4) Statement-1 is false, Statement-2 is false.





15. If a_1, a_2, a_3 ($a_i > 0$) are in G.P. with common ratio r , then the value of r for which the inequality $9a_1 + 5a_3 > 14a_2$ holds cannot lie in the interval
- 1) $(0,1)$ 2) $\left[1, \frac{9}{5}\right]$ 3) $\left(\frac{4}{5}, 1\right)$ 4) $\left[\frac{5}{9}, 1\right)$
16. Which of the following statements is/are correct ?
- A: For some natural N , the number of positive integral 'x' satisfying the equation, $1! + 2! + 3! + \dots + (x!) = (N)^2$ is 3
- B: A women has 11 close friends then the number of ways in which she can invite 5 of them to dinner, if two particular of them are not on speaking terms & will not attend together is 378.
- C: An old man while dialing a 7 digit telephone number remembers that the first four digits consists of one 1's, one 2's and two 3's. He also remembers that the fifth digit is either a 4 or 5 and while he doesn't remember the sixth digit, he remembers that the seventh digit is 9 minus the sixth digit. Maximum number of distinct trials he has to try to make sure that he dials the correct telephone number, is 340.
- D: The number of times the digit 3 will be written when listing the integers from 1 to 1000 is 300.
- 1)ABC 2)BC 3)BCD 4)BD
17. Consider the binomial distribution, $B(n=16, p)$. The mean and standard deviation of the binomial variate X are 4 and 3 respectively. Then $P(X \geq 1) =$
- 1) $1 - \left(\frac{1}{4}\right)^{16}$ 2) $1 - \left(\frac{3}{4}\right)^{16}$ 3) $1 - \left(\frac{2}{3}\right)^{16}$ 4) $1 - \left(\frac{1}{3}\right)^{16}$
18. Let $\Delta(x) = \begin{vmatrix} \cos 2x & \sin^2 x & \cos 4x \\ \sin^2 x & \cos 2x & \cos^2 x \\ \cos 4x & \cos^2 x & \cos 2x \end{vmatrix} = a_0 + a_1 \sin x + a_2 \sin^2 x + \dots, (\forall x \in R)$, then the value of $a_1 + a_2 + 18$ is equals to
- 1)0 2)18 3)17 4)5





19. Let $Z_r, r = 1, 2, \dots, n-1, n$, be roots of equation ${}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n = 0$ in Argand's plane.

If there exists exactly one Z_r such that $\arg\left(\frac{Z_r - (-1 + \sqrt{2}i)}{(-1) - (-1 + \sqrt{2}i)}\right) = \frac{\pi}{4}$, then n can be

1) 10

2) 12

3) 14

4) 8

20. If $P_1: \vec{r} \cdot \vec{n}_1 - d_1 = 0$, $P_2: \vec{r} \cdot \vec{n}_2 - d_2 = 0$ and $P_3: \vec{r} \cdot \vec{n}_3 - d_3 = 0$ are three planes and $\vec{n}_1, \vec{n}_2$ and $\vec{n}_3$ are three non-coplanar vectors then, the three lines $P_1 = P_2 = 0$, $P_2 = P_3 = 0$ and $P_3 = P_1 = 0$ are

1) parallel lines

2) coplanar lines

3) coincident lines

4) concurrent lines

SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$, respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \leq d_1(P) + d_2(P) \leq 4$, is

$$22. \text{ If } f(x) = \begin{cases} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1 - \{x\}^2)\right) \sin^{-1}(1 - \{x\})}{\sqrt{2}(\{x\} - \{x\}^3)}, & x > 0 \\ k, & x = 0 \\ \frac{A \sin^{-1}(1 - \{x\}) \cos^{-1}(1 - \{x\})}{\sqrt{2}\{x\}(1 - \{x\})}, & x < 0 \end{cases}$$

A^2 is _____. (Where $\{x\}$ denotes fractional part of x).

23. Let f and g be continuously differentiable functions such that $f(0) = 0, f'(0) = 2$ and $g(x) = f(-x + f(f(x)))$. The value of $g'(0)$ equals

24. Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a continuous strictly increasing function, such that

$$f^3(x) = \int_0^x t \cdot f^2(t) dt \text{ for every } x \geq 0, \text{ then value of } f(6) \text{ is } \underline{\hspace{2cm}}$$

25. If $\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{{}^nC_k}{n^k(k+3)} = p$ then $[p+1]$ is _____ (where $[.]$ denotes greatest integer function)

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PHYSICS

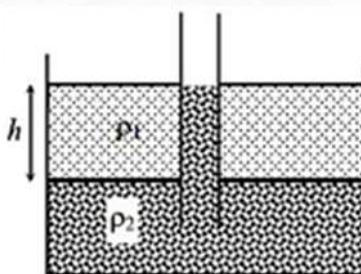
Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- 26.** If $\vec{L}_{CM}$ is the angular momentum about centre of mass of a system of particles , $\vec{r}_{cm}$ is the position vector of the centre of mass relative to any point ‘A’ and $\vec{p}$ is the total momentum, then the angular momentum of the system relative to the point A is given by
- 1) $\vec{L} = \vec{L}_{CM}$ 2) $\vec{L} = \vec{L}_{CM} + \vec{p} \times \vec{r}_{cm}$
- 3) $\vec{L} = \vec{L}_{CM} + \vec{r}_{cm} \times \vec{p}$ 4) $\vec{L}_{CM} = \vec{L} + \vec{r}_{cm} \times \vec{p}$
- 27.** A container has two immiscible liquids of densities ρ_1 and $\rho_2 (> \rho_1)$. A capillary tube of radius r is inserted in the liquid so that its bottom reaches upto the denser liquid. The denser liquid rises in the capillary and attains a height h from interface of the liquids, which is equal to the column length of the lighter liquid. Assuming angle of contact to be zero, the surface tension of heavier liquid is



28. The kinetic energy K of a particle moving along a circle of radius R depends upon the distance moved from rest s , as $K = as^2$. The force acting on the particle is (a is a positive constant)
- 1) $2a\pi\rho_2gh$ 2) $\frac{\rho_2 rgh}{2}$ 3) $\frac{r}{2}(\rho_2 - \rho_1)gh$ 4) $2\pi r(\rho_2 - \rho_1)gh$
- 1) $2a\frac{s^2}{R}$ 2) $2as\left[1 + \frac{s^2}{R^2}\right]^{\frac{1}{2}}$ 3) $2as$ 4) $2a$

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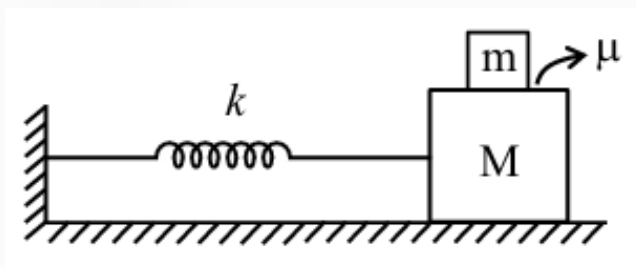
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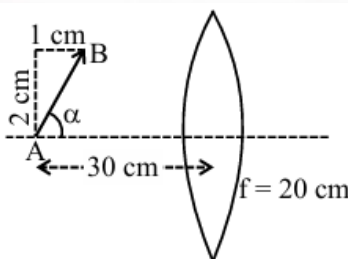
29. Two blocks of masses m and M , ($M > m$), are placed on a frictionless table as shown in figure. A massless spring with spring constant k is attached with the lower block. If the system is slightly displaced and released then the two blocks execute SHM together (μ = coefficient of friction between the two blocks)



- (A) The time period of small oscillation of the two blocks is $T = 2\pi\sqrt{\frac{(m+M)}{k}}$
- (B) The magnitude of acceleration of the blocks is $a = \frac{kx}{M+m}$ (x = displacement of the blocks from the mean position)
- (C) The magnitude of the frictional force on the upper block is $\frac{m\mu|x|}{M+m}$
- (D) The maximum amplitude of the upper block, if it does not slip, is $\frac{\mu(M+m)g}{k}$
- (E) Maximum frictional force can be $\mu(M+m)g$.

Choose the correct answer from the options given below :

- 1) A, B, D Only 2) B, C, D Only 3) C, D, E Only 4) A, B, C Only
30. A slanted object AB is placed on one side of convex lens as shown in the diagram. The image is formed on the opposite side. Angle made by the image with principal axis is :

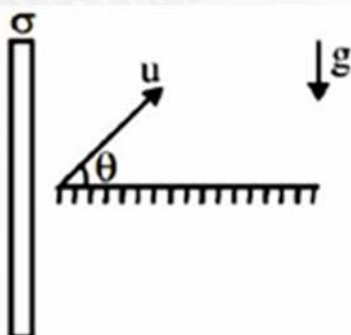


- 1) $-\frac{\alpha}{2}$ 2) -45° 3) $+45^\circ$ 4) $-\alpha$





31. A close organ pipe of diameter 10 cm has length 42 cm. The air column in pipe vibrates in its second overtone with the maximum pressure amplitude ΔP_0 . The pressure amplitude at middle of pipe is
- 1) $\frac{\Delta P_0}{2}$ 2) $\frac{\Delta P_0}{\sqrt{2}}$ 3) $\frac{\sqrt{3}\Delta P_0}{2}$ 4) ΔP_0
32. The molar heat capacity of an ideal gas in a process varies as $C = C_V + \alpha T^2$ (where C_V is molar heat capacity at constant volume and α is a constant) Then the equation of the process is
- 1) $V e^{\left(\frac{\alpha T^2}{2R}\right)} = \text{constant}$ 2) $V e^{\left(\frac{\alpha T^2}{R}\right)} = \text{constant}$
- 3) $V e^{\left(\frac{2\alpha T^2}{R}\right)} = \text{constant}$ 4) $V e^{\left(\frac{3\alpha T^2}{2R}\right)} = \text{constant}$
33. In the set-up shown, there is a large thin non-conducting vertical sheet with charge density σ . In the nearby region a charge q of mass m is projected at an angle θ with initial velocity u . When this charge reaches its maximum height find its velocity.

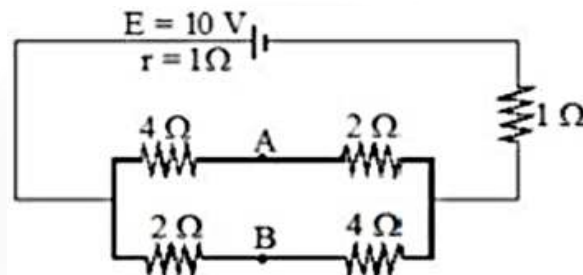


- 1) $u \cos \theta \left(1 + \frac{\sigma q \tan \theta}{2m\epsilon_0 g}\right)$ 2) $u \cos \theta \left(1 + \frac{\sigma q \tan \theta}{m\epsilon_0 g}\right)$
- 3) $u \sin \theta \left(1 + \frac{\sigma q \tan \theta}{2m\epsilon_0 g}\right)$ 4) $u \sin \theta \left(1 + \frac{\sigma q \tan \theta}{m\epsilon_0 g}\right)$
34. A uniform wound solenoidal coil of self-inductance $1.8 \times 10^{-4} \text{ H}$ and resistance 6Ω is broken up into two identical coils. These identical coils are then connected in parallel across a 12V battery of negligible resistance. The time constant of the circuit is
- 1) $3 \times 10^{-5} \text{ s}$ 2) $1.5 \times 10^{-5} \text{ s}$ 3) $0.75 \times 10^{-5} \text{ s}$ 4) $6 \times 10^{-5} \text{ s}$





35. In the circuit shown below, the cell has an emf of 10 V and internal resistance of 1 ohm. The other resistances are shown in the figure. The potential difference $V_A - V_B$ is



- 1) -4V 2) 4V 3) 2V 4) -2V
36. Considering Bohr's atomic model for hydrogen atom :
- (A) The energy of H atom in ground state is same as energy of He^+ ion in its first excited state.
- (B) The energy of H atom in ground state is same as that for Li^{++} ion in its second excited state.
- (C) The energy of H atom in its ground state is same as that of He^+ ion for its ground state.
- (D) The energy of He^+ ion in its first excited state is same as that for Li^{++} ion in its ground state
- choose the correct answer from the options given below:
- 1) (B), (D) only 2) (A), (B) only 3) (A), (D) only 4) (A), (C) only
37. **Statement-1:** Net dipole moment of a polar linear isotropic dielectric substance is not zero even in the absence of an external electric field.
- Statement-2:** In absence of an external electric field, the different permanent dipoles of a polar dielectric substance are oriented in random directions.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- 1) **Statement-1** is correct but **Statement-2** is not correct
- 2) Both **Statement-1&2** are correct but **Statement-2** is not the correct explanation of **Statement-1**
- 3) Both **Statement-1&2** are correct but **Statement-2** is the correct explanation of **Statement-1**
- 4) **Statement-1** is not correct but **Statement-2** is correct





38. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: In photoelectric effect, on increasing the intensity of incident light the stopping potential increases.

Reason R: Increase in intensity of light increases the rate of photoelectrons emitted, provided the frequency of incident light is greater than threshold frequency.

In the light of the above statements, choose the correct answer from the options given below

- 1) Both A and R are true but R is NOT the correct explanation of A
 - 2) A is false but R is true
 - 3) A is true but R is false
 - 4) Both A and R are true and R is the correct explanation of A
39. **Statement-1:** The kinetic energy needed to project a body of mass m from earth surface to infinity is $\frac{1}{2} mgR$, where R is the radius of earth.
- Statement-2:** The maximum potential energy of a body is zero when it is projected to infinity from earth surface.
- In the light of the above statements, choose the correct answer from the option given below
- 1) **Statement-1** is not correct but **Statement-2** is correct
 - 2) Both **Statement-1&2** are correct but **Statement-2** is the correct explanation of **Statement-1**
 - 3) **Statement-1** is correct but **Statement-2** is not correct
 - 4) Both **Statement-1&2** are correct but **Statement-2** is not the correct explanation of **Statement-1**
40. Consider a n-type semiconductor in which n_e and n_h are number of electrons and holes, respectively.
- (A) Holes are minority carriers
 - (B) The dopant is a pentavalent atom
 - (C) $n_e n_h \neq n_i^2$ (where n_i is number of electrons or holes in semiconductor when it is intrinsic form)
 - (D) $n_e n_h \geq n_i^2$
 - (E) The holes are not generated due to the donors
- Choose the correct answer from the options given below:
- 1) (A), (C), (D) only
 - 2) (A), (C), (E) only
 - 3) (A), (B), (E) only
 - 4) (A), (B), (C) only





41. **Assertion (A):** In Young's double slit experiment, the fringes produced by red light are closer as compared to those produced by blue light.

Reason (R): The fringe width is directly proportional to the wavelength of light. In the light of above statements, choose the correct answer from the options given below:

- 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- 2) (A) is false but (R) is true.
- 3) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- 4) (A) is true but (R) is false.

42. List-I shows various functional dependencies of energy E on the atomic number (Z).

Energies associated with certain phenomena are given in List-II

Choose the option that describes the correct match between the entries in List-I to those in List-II

	List-I		List-II
P)	$E \propto Z^2$	1)	Energy of characteristic x-rays
Q)	$E \propto (Z-1)^2$	2)	Electrostatic part of the nuclear binding energy for stable nuclei with mass numbers in the range 30 to 170
R)	$E \propto Z(Z-1)$	3)	Energy of continuous x-rays
S)	E is practically independent of Z	4)	Average nuclear binding energy per nucleon for stable nuclei with mass number in the range 30 to 170
		5)	Energy of radiation due to electronic transitions from hydrogen-like atoms

- 1) $(P) \rightarrow (4), (Q) \rightarrow (3), (R) \rightarrow (1), (S) \rightarrow (2)$
- 2) $(P) \rightarrow (5), (Q) \rightarrow (2), (R) \rightarrow (1), (S) \rightarrow (4)$
- 3) $(P) \rightarrow (5), (Q) \rightarrow (1), (R) \rightarrow (2), (S) \rightarrow (4)$
- 4) $(P) \rightarrow (3), (Q) \rightarrow (2), (R) \rightarrow (1), (S) \rightarrow (5)$

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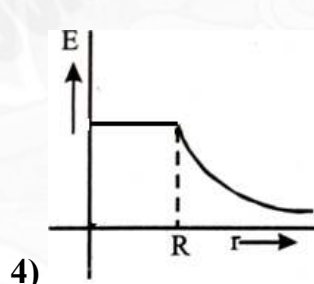
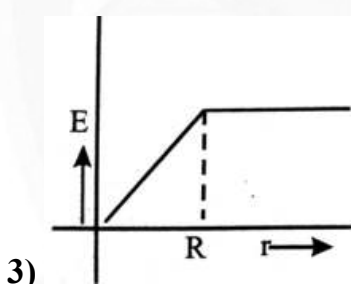
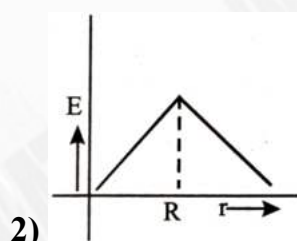
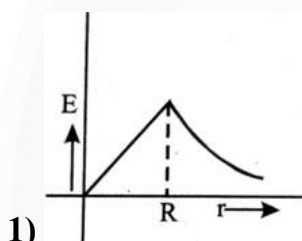
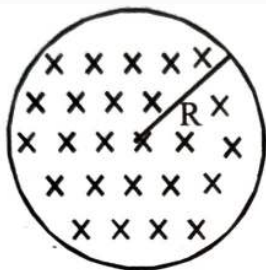
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43. Figure shows a circular area of radius R where a uniform magnetic fields $\vec{B}$ is going into the plane of paper and increasing in magnitude at a constant rate. In that case, which of the following graphs, drawn schematically, correctly shows the variation of the induced electric field $E(r)$



44. In a certain series LCR circuit consisting of an AC source with an emf given by $V = V_m \sin \omega t$, it is found that the inductive reactance is twice the capacitive reactance ($X_L = 2X_C$) and phase difference between the current and source voltage is $\frac{\pi}{4}$. If the capacitance is now made one-quarter ($1/4$) times its original value, and the self-inductance and resistance are both doubled, which of the following statement is correct?
- 1) Source voltage will lead current by a phase of $V_m \sin \omega t$,
 - 2) Source voltage will lag behind current by a phase of $\tan^{-1}\left(\frac{1}{2}\right)$
 - 3) Source voltage will lag behind current by a phase of $\frac{\pi}{4}$
 - 4) Source voltage and current will be in same phase



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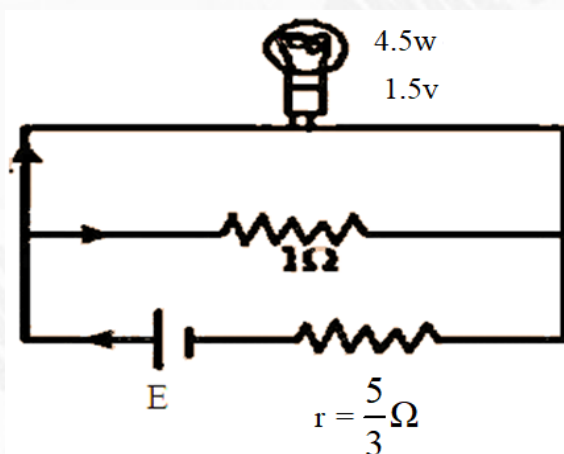
45. A cylindrical filament with a length of 10 cm and a diameter of 0.2 mm, having an emissivity of 0.2, is at a temperature of 2000K inside an evacuated light bulb. Calculate the power (In watts) radiated from its curved surface. (Take $\sigma = 5.67 \times 10^{-8} \text{ W / m}^2 \text{ K}^4$)
- 1) 11.4 2) 1140 3) 114 4) 1.14

SECTION-II
(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. A Torch bulb rated at 4.5 W, 1.5 V is connected as shown in the figure, to make the bulb glow at full intensity, what should be the emf (in volt) of the cell?

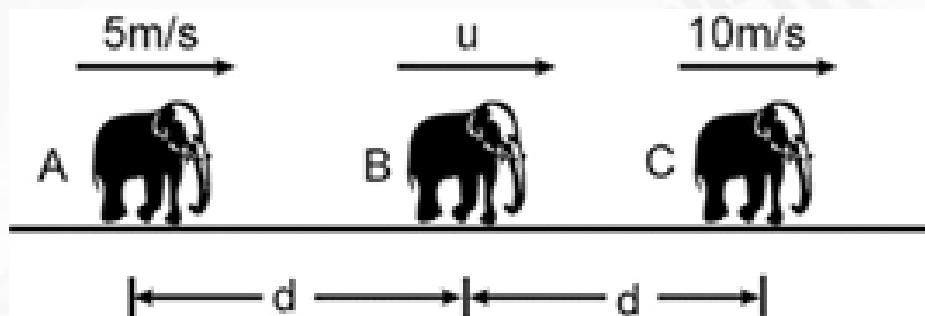


47. A train with cross-sectional area S_t is moving with speed v_t inside a long tunnel of cross-sectional area s_0 ($s_0 = 3s_t$). Assume that almost all the air (density ρ) in front of the train flows back between its sides and the walls of the tunnel. Also, the air flow with respect to the train is steady and laminar. Take the ambient pressure and that inside the train to be p_0 . If the pressure in the region between the sides of the train and the tunnel walls is p , then $p_0 - p = \frac{5}{2N} \rho v_t^2$. The value of N is.....

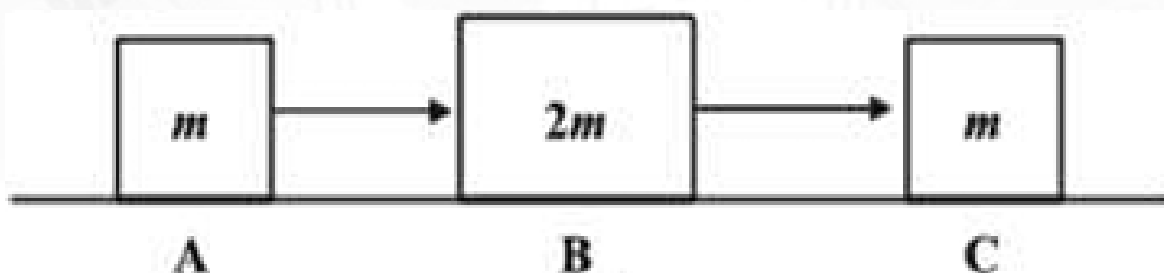




48. The ratio of the mean free paths of the molecules of two gases having molecular diameters $1 \text{ \AA}$ and $2 \text{ \AA}$ is $x:1$, find x . The gases may be considered under identical conditions of temperature, pressure and volume.
49. Three elephants A, B and C are moving along a straight line with constant speed in same direction as shown in figure. Speed of A is 5 m/s and speed of C is 10 m/s . Initially separation between A & B is ' d ' and between B & C is also d . When 'B' catches 'C' separation between A & C becomes $3d$. Find the speed of B (in m/s)



50. Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. These have masses m , $2m$ and m , respectively. The object A moves towards B with a speed 9 m/s and makes an elastic collision with it. Thereafter B makes completely inelastic collision with C. All motions occur on the same straight line. Find the final speed (in m/s) of the object C (perfectly inelastic collision $e=0$)



**CHEMISTRY****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. For any given series of spectral lines of atomic hydrogen, let $\Delta \bar{\nu} = \bar{\nu}_{\max} - \bar{\nu}_{\min}$ be the difference in maximum and minimum frequencies in cm^{-1} . The ratio $\Delta \bar{\nu}_{\text{Lyman}} / \Delta \bar{\nu}_{\text{Balmer}}$ is
- 1) 4:1 2) 27:5 3) 9:4 4) 5:4

52. **Statement I:** Dipole moment is a vector quantity and by convention it is depicted by a small arrow with tail on the negative centre and head pointing towards the positive centre.

Statement II: The crossed arrow of the dipole moment symbolize the direction of the shift of charges in the molecules.

In the light of the above statements, choose the most appropriate answer from the options given below:

- 1) Both statement I and Statement II are correct.
 2) Statement I is correct but statement II is incorrect
 3) Statement I is incorrect but Statement II is correct
 4) Both Statement I and Statement II are incorrect

53. $A_{(g)} \rightleftharpoons B_{(g)} + \frac{C_{(g)}}{2}$

The correct relationship between K_p , α and equilibrium pressure P is

- 1) $K_p = \frac{\alpha^{1/2} P^{1/2}}{(2+\alpha)^{3/2}}$ 2) $K_p = \frac{\alpha^{1/2} P^{1/2}}{(2+\alpha)^{1/2}}$
 3) $K_p = \frac{\alpha^{1/2} P^{3/2}}{(2+\alpha)^{3/2}}$ 4) $K_p = \frac{\alpha^{3/2} P^{1/2}}{(2+\alpha)^{1/2} (1-\alpha)}$

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54. Calculate the standard cell potential (inV) of the cell in which following reaction takes place. $\text{Fe}^{2+}_{(\text{aq})} + \text{Ag}^{+}_{(\text{aq})} \rightleftharpoons \text{Fe}^{3+}_{(\text{aq})} + \text{Ag}_{(\text{s})}$ Given that

$$E^0_{\text{Ag}^+/\text{Ag}} = x\text{V}; E^0_{\text{Fe}^{2+}/\text{Fe}} = y\text{V}; E^0_{\text{Fe}^{3+}/\text{Fe}} = z\text{V}$$

- 1) $x - z$ 2) $x - y$ 3) $x + 2y - 3z$ 4) $x + y - z$
55. The electronic configuration for neodymium is [Atomic number for neodymium 60]
- 1) $[\text{Xe}]5f^7 7s^2$ 2) $[\text{Xe}]4f^6 6s^2$ 3) $[\text{Xe}]4f^1 5d^1 6s^2$ 4) $[\text{Xe}]4f^4 6s^2$
56. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A): H_2Te is more acidic than H_2S .

Reason (R): Bond dissociation enthalpy of H_2Te is lower than H_2S .

In the light of the above statements, choose the most appropriate from the options given below.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true but R is not the correct explanation of A.
- 3) A is false but R is true.
- 4) A is true but R is false.
57. $\text{A} \xrightarrow{4\text{KOH}, \text{O}_2} \underset{(\text{green})}{2\text{B}} + 2\text{H}_2\text{O}$
- $3\text{B} \xrightarrow{4\text{HCl}} \underset{(\text{Purple})}{2\text{C}} + \text{MnO}_2 + 2\text{H}_2\text{O}$
- $2\text{C} \xrightarrow{\text{H}_2\text{O}, \text{KI}} 2\text{A} + 2\text{KOH} + \text{D}$ in the above sequence of reactions, A and D respectively are
- 1) KI and KMnO_4 2) KIO_3 and MnO_2
- 3) KI and K_2MnO_4 4) MnO_2 and KIO_3





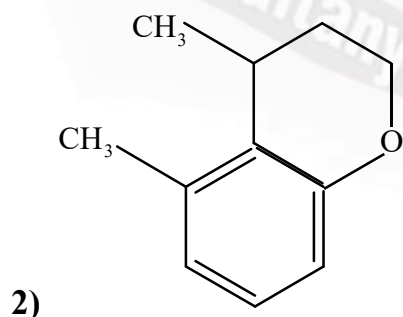
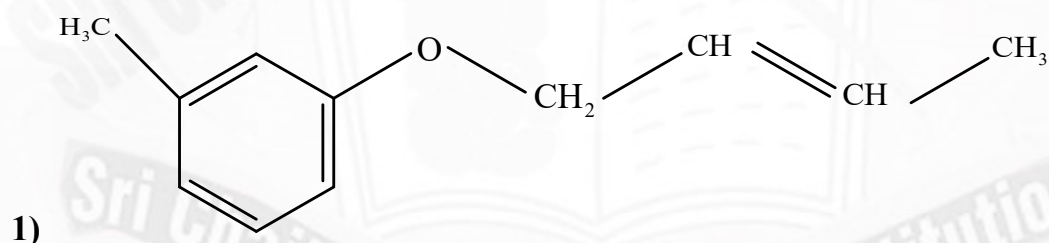
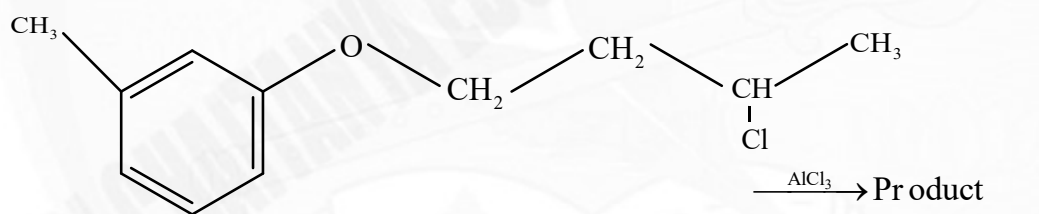
58. Match List I with List II

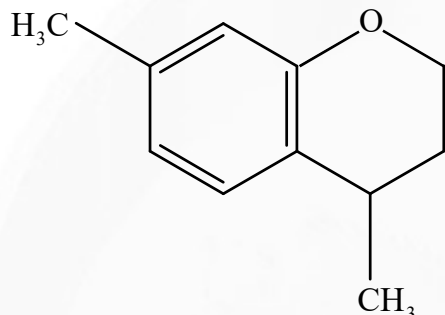
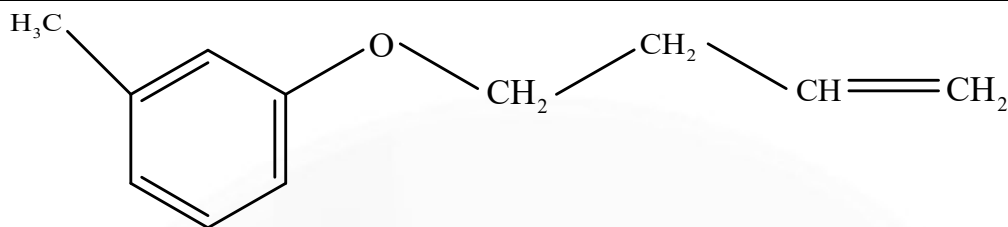
	List I (Complex)		List II [Crystal Field Splitting Energy (Δ_0)]
A	$[\text{Ti}(\text{H}_2\text{O})_6]^{2+}$	I	-1.2
B	$[\text{V}(\text{H}_2\text{O})_6]^{2+}$	II	-0.6
C	$[\text{Mn}(\text{H}_2\text{O})_6]^{3+}$	III	0
D	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	IV	-0.8

Choose the correct answer from the options given below:

- 1) A – IV, B – I, C – III, D – II 2) A – II, B – IV, C – III, D – I
 3) A – II, B – IV, C – I, D – III 4) A – IV, B – I, C – II, D – III

59. The major product obtained in the given reaction is





60. Given below are two statements:

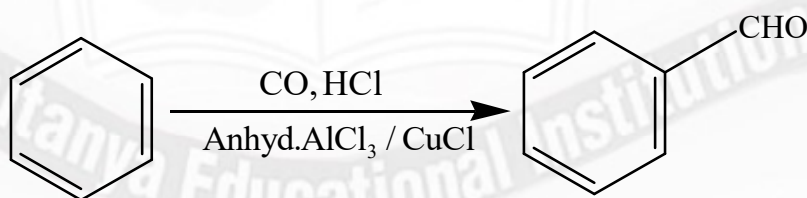
Statement I: On heating with KHSO_4 , glycerol is dehydrated and acrolein is formed.

Statement II: Acrolein has fruity odour and can be used to test glycerol's presence.

Choose the correct option.

- 1) Both statement I and statement II are correct
- 2) Both statement I and Statement II are incorrect.
- 3) Statement I is correct but statement II is incorrect.
- 4) Statement I is incorrect but statement II is correct.

61. Identify the name of the reaction

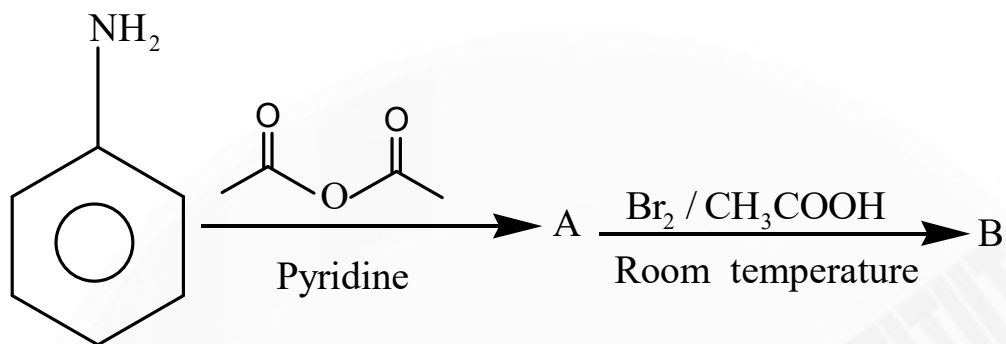


- | | |
|------------------------|----------------------------|
| 1) Stephen reaction | 2) Etard reaction |
| 3) Rosenmund reduction | 4) Gatterman-Koch reaction |





62. The major products A and B formed in the following reaction sequence are

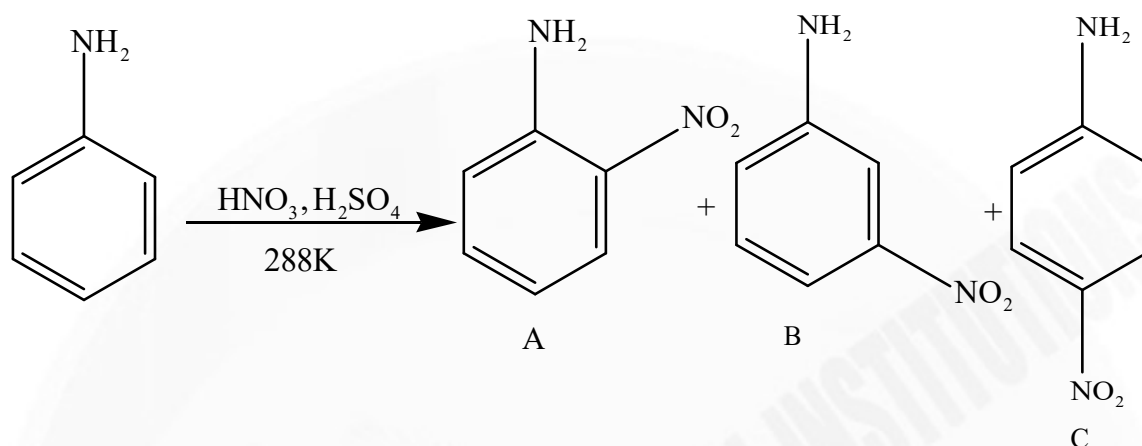


- Options for products A and B:
- 1) A = CC(=O)c1ccc(N)cc1 (p-Acetanilide), B = CC(=O)c1ccc(N)c(Br)c1 (3-bromo-p-acetanilide)
- 2) A = CC(=O)c1ccc(N)cc1 (p-Acetanilide), B = CC(=O)Nc1ccccc1Br (o-bromoacetanilide)
- 3) A = CC(=O)Nc1ccccc1 (Acetanilide), B = CC(=O)Nc1ccc(Br)cc1 (p-bromoacetanilide)
- 4) A = CC(=O)c1ccc(N)cc1 (p-Acetanilide), B = CC(=O)c1c(Br)cc(N)c(Br)c1 (2,6-dibromo-p-acetanilide)





63. Consider the given reaction, percentage yield of



- 1) $C > A > B$ 2) $C > B > A$ 3) $A > C > B$ 4) $B > C > A$

64. Given below are two statements one is labelled as assertion and other is labelled as Reason.

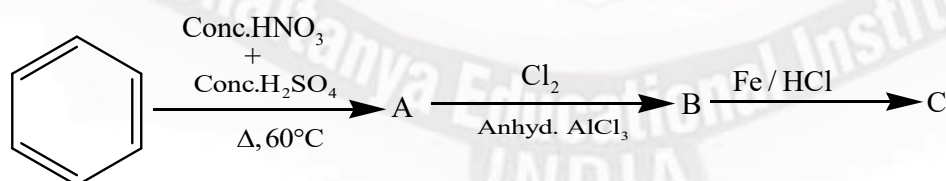
Assertion(A): Gabriel phthalimide synthesis cannot be used to prepare aromatic primary amines.

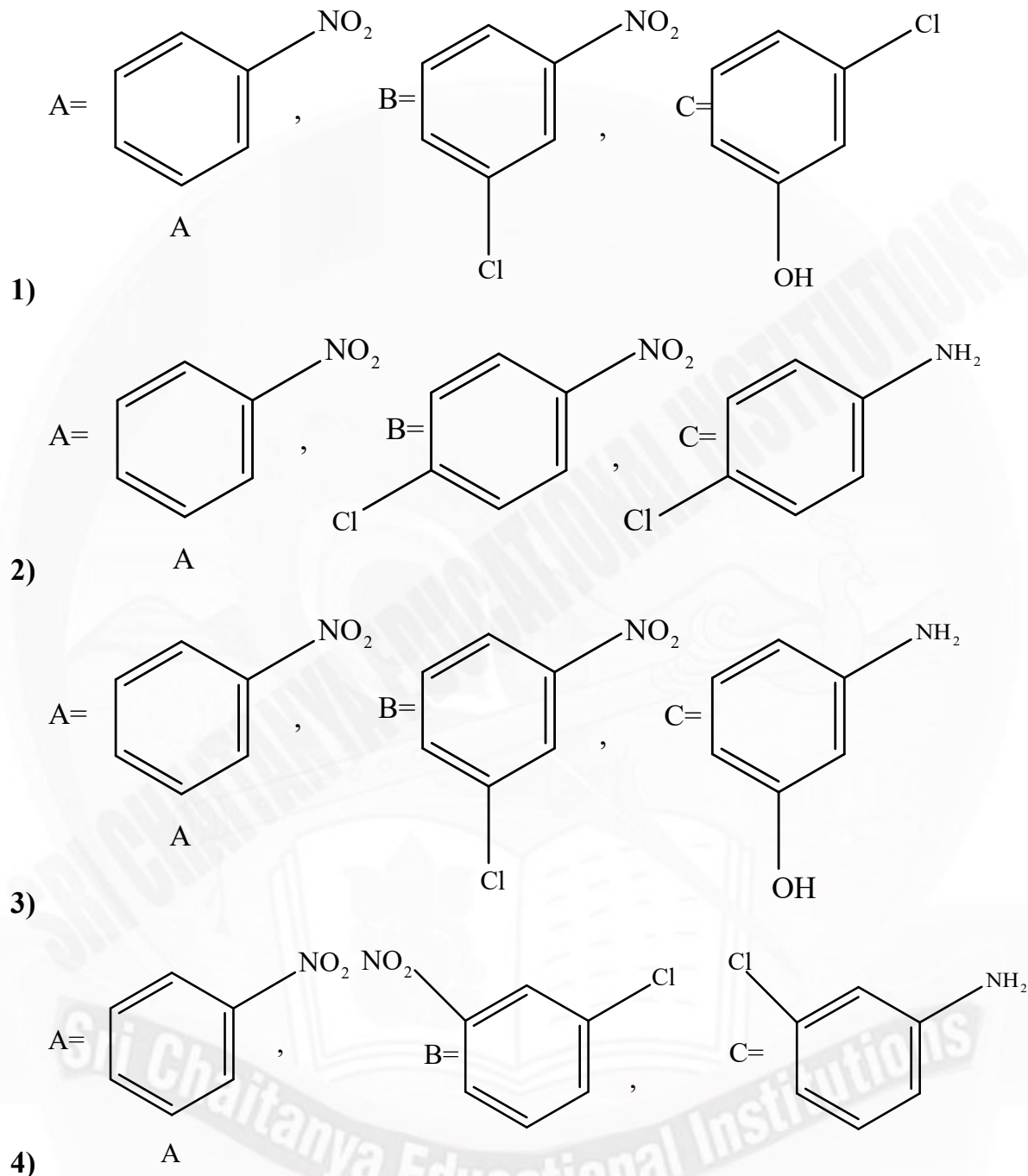
Reason(R): Aryl halides do not undergo nucleophilic substitution reactions.

In the light of the above statements, choose the correct answer from options given below

- 1) Both A and R are true and R is the correct explanation of A.
 2) Both A and R are true but R is not the correct explanation of A.
 3) A is false but R is true.
 4) A is true but R is false.

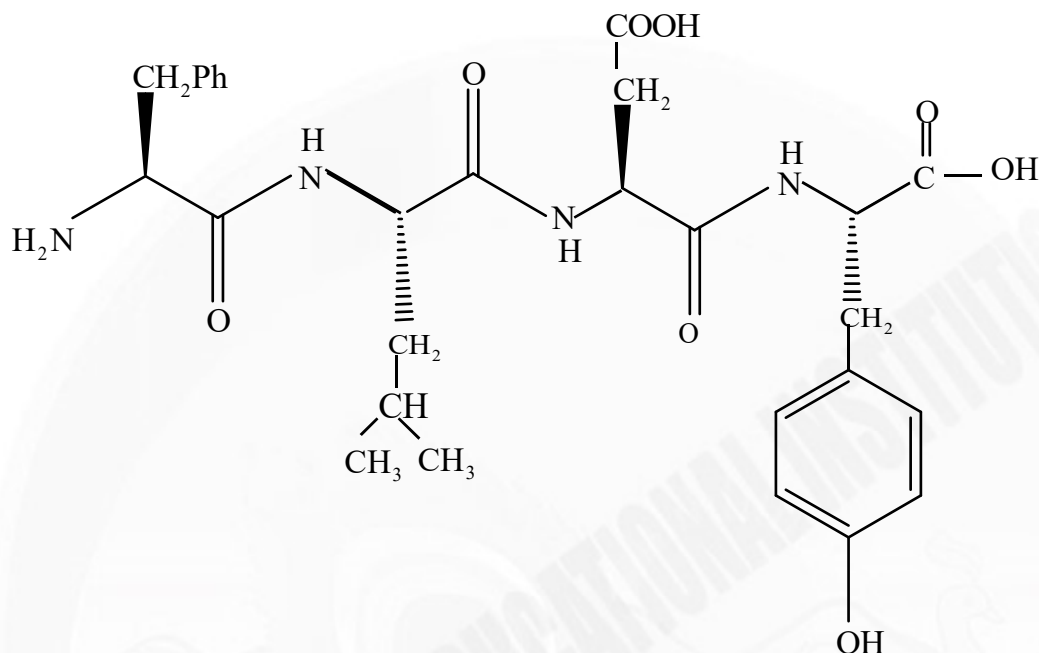
65. Identify correct A, B and C in the reaction sequence given below.







66. Following tetra peptide can be represented as



(F,L,D,Y,I,Q,P are one letter codes for amino acids.)

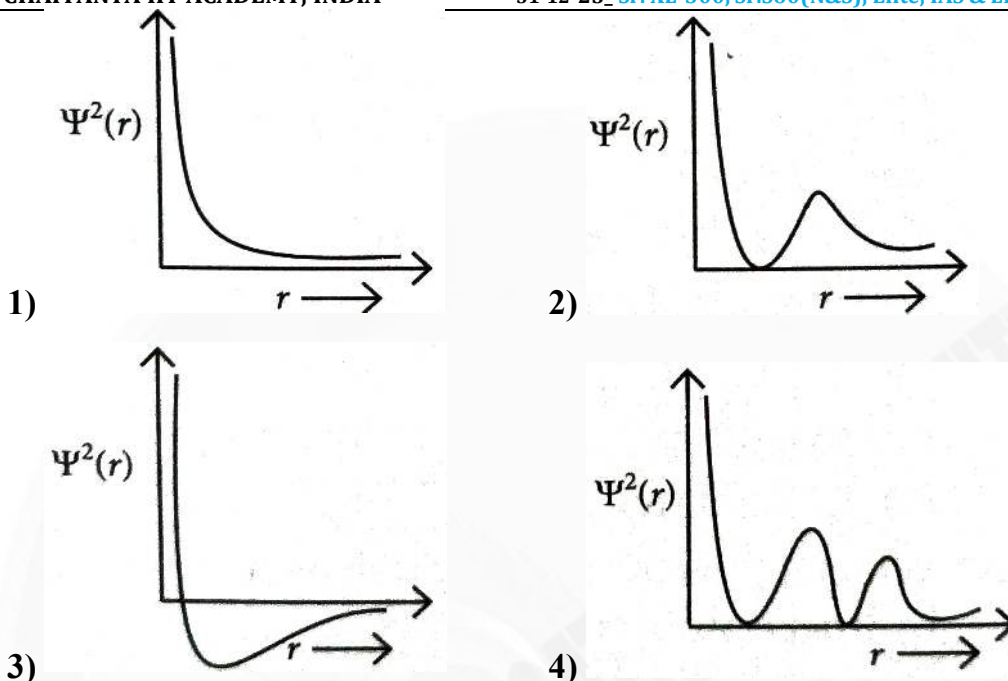
- 1) FIQY 2) YQLF 3) PLDY 4) FLDY

67. Complex A has a composition of $\text{H}_{12}\text{O}_6\text{Cl}_3\text{Cr}$. If the complex on treatment with conc. H_2SO_4 loses 13.5% of its original mass, the correct molecular formula of A is [Given: atomic mass Cr = 52 amu and Cl = 35 amu]

- 1) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ 2) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
 3) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ 4) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$

68. Which of the following is the correct plot for the probability density $\psi^2(r)$ as a function of distance 'r' of the electron from the nucleus for 2s orbital?





69. The values of molar conductivity for different concentrations for KCl in aqueous Solution are as shown

$\wedge_M (S - cm^2 / mol)$	200	150	100
$C (mol / lit)$	0.0025	0.01	0.0225

Calculate the molar conductivity at infinite dilution for KCl solution in $S - cm^2 / mol$

- 1) 300 2) 250 3) 216 4) 225
70. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of this solution at 298 K is $2.0 g cm^{-3}$. The ratio of the molecular weights of the solute and solvent, $\left(\frac{MW_{solute}}{MW_{solvent}} \right)$, is?
- 1) 9 2) 18 3) 27 4) 36

SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

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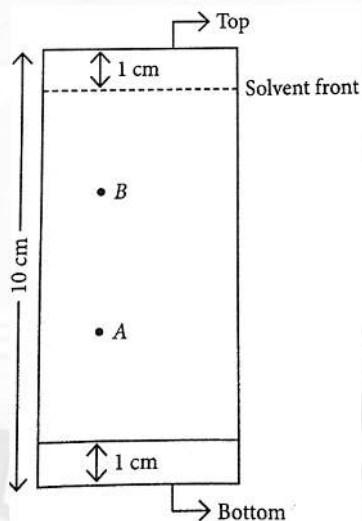
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71. 15 ml of aqueous solution of Fe^{2+} in acidic medium completely reacted with 20 mL of 0.03 M aqueous $\text{Cr}_2\text{O}_7^{2-}$. The molarity of the Fe^{2+} solution is $___ \times 10^{-2}$ M.
72. The standard entropy change for the reaction: $4\text{Fe}_{(s)} + 3\text{O}_{2(g)} \rightarrow 2\text{Fe}_2\text{O}_{3(s)}$ is -550 JK^{-1} at 298 K. [Given: The standard enthalpy changes for the reaction is -165 kJ mol^{-1}]
The temperature in K at which the reaction attains equilibrium is $___$ (Nearest integer)
73. Catalyst A reduces the activation energy for a reaction by 10 kJ mol^{-1} at 300 K. The ratio of rate constants $\frac{K_{T, \text{catalysed}}}{K_{T, \text{uncatalysed}}}$ is e^x . The value of x is $___$ (Nearest integer)
[Assume that the pre-exponential factor is same in both the cases. Given, $R = 8.31 \text{ JK}^{-1} \text{ mol}^{-1}$]
74. In the given TLC, the distance of spot A and B are 5 cm and 7 cm, from the bottom of TLC plate, respectively.



Retardation factor (R_f) value of B is $x \times 10^{-1}$ times more than A. The value of x is $___$

75. In the Claisen-Schmidt reaction to prepare 351 g of dibenzal acetone using 87g of acetone, the amount of benzaldehyde required is $___ \text{ g}$. (nearest integer)





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500
ALL INDIA OPEN CATEGORY RANKS

113

BELOW
1000
ALL INDIA OPEN CATEGORY RANKS

205

BELOW
1000
ALL INDIA CATEGORY RANKS COUNT

745

**NUMBER OF
QUALIFIED RANKS**
4,212

*DLP/AITS



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